EXPERIMENT 6

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Experiment 6 – Understanding and Implementing Cursors for Row-by-Row Data Processing (Oracle, Accenture, Scaler, D. E. Shaw & Co)

Experiment

Experiment 6: Working with PL/SQL cursors to process multiple rows of a table individually. This experiment demonstrates the use of implicit cursors, explicit cursors, and cursor attributes to implement business logic on row-by-row data in Oracle and PostgreSQL.

Aim

To understand the concept and working of cursors in PL/SQL for row-by-row data processing, and to analyze how implicit cursors, explicit cursors, and cursor attributes are used to implement business logic on multiple rows in a database table.

Objective

- To implement and analyze implicit cursors for single-row operations.
- To implement explicit cursors to fetch and process multiple records.
- To apply cursor attributes (%FOUND, %NOTFOUND, %ROWCOUNT, %ISOPEN) for controlling program flow.
- To process employee records row by row and apply business logic effectively.

Software Requirements

Database Management System:

- Oracle Database Express Edition (Oracle XE)

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- PostgreSQL Database

Database Administration Tool / Client Tool:

- Oracle SQL Developer (for Oracle XE)
- pgAdmin (for PostgreSQL)

Problem Statement

In real-world enterprise applications, database queries often return multiple rows that need to be processed individually to apply specific business rules. Using cursors allows row-by-row processing for detailed business logic implementation.

Practical / Experiment Steps

1. Design PL/SQL programs that demonstrate:
    - The use of implicit cursors for single-row DML operations
    - The use of explicit cursors to fetch and process multiple records
    - The application of cursor attributes to control program execution
2. Create PL/SQL programs to:
    - Fetch employee records from a database table using cursors
    - Process each record individually
    - Display results or apply business logic using cursor attributes

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Procedure

1. Open Oracle SQL Developer or pgAdmin and connect to the database.
2. Create an employee table with sample data (id, name, department, salary).
3. Write an anonymous PL/SQL block demonstrating implicit cursor operations.
4. Write a PL/SQL block using an explicit cursor to fetch and process multiple records.
5. Apply cursor attributes such as %FOUND, %NOTFOUND, %ROWCOUNT, %ISOPEN to control execution flow.

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6. Execute the blocks and observe the output.
7. Capture results and screenshots for documentation.

Input / Output Details

Input

employee table:

- id (Integer)
- name (Varchar)
- department (Varchar)
- salary (Integer)

Cursor logic used:

- Implicit cursor for single-row processing
- Explicit cursor for multi-row processing
- Cursor attributes (%FOUND, %NOTFOUND, %ROWCOUNT, %ISOPEN)

Step-wise Output

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Step 1 – Create employee Table

```
DELETE FROM EMPLOYEE
WHERE
        ID = :ID
    AND NAME = :NAME
    AND DEPARTMENT = :DEPARTMENT
    AND SALARY = :SALARY;

DESCRIBE EMPLOYEE;

DELETE FROM EMPLOYEE
WHERE
        ID = :ID
    AND NAME = :NAME
    AND DEPARTMENT = :DEPARTMENT
    AND SALARY = :SALARY;

DELETE FROM STUDENTS
WHERE
        NAME = :NAME
    AND MARKS = :MARKS;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
PL/SQL procedure successfully completed.

Elapsed: 00:00:00.079
```

```
SQL> DECLARE
        -- Declare explicit cursor
        CURSOR emp_cursor IS
            SELECT id, name, department, salary FROM employee;...
Show more...
```

Step 2 – Insert Data into employee Table

```
-- Step 1 - Create employee table
CREATE TABLE employee (
    id NUMBER PRIMARY KEY,
    name VARCHAR2(50),
    department VARCHAR2(50),
    salary NUMBER
);
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
Table EMPLOYEE created.

Elapsed: 00:00:00.014
```

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Step 3 – Implicit Cursor Operation

```sql
-- Step 2 - Insert sample employee data
INSERT INTO employee VALUES (1, 'Rahul', 'HR', 25001);
INSERT INTO employee VALUES (2, 'Sneha', 'Finance', 32000);
INSERT INTO employee VALUES (3, 'Amit', 'IT', 45015);
INSERT INTO employee VALUES (4, 'Neha', 'Marketing', 28000);
INSERT INTO employee VALUES (5, 'Vikram', 'Sales', 39011);

COMMIT;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> INSERT INTO employee VALUES (2, 'Sneha', 'Finance', 32000)


1 row inserted.

Elapsed: 00:00:00.002
```

Step 4 – Explicit Cursor Declaration and Fetch

```sql
DECLARE
    v_salary employee.salary%TYPE;
BEGIN
    -- Implicit cursor: single row query
    SELECT salary INTO v_salary
    FROM employee
    WHERE id = 2;

    DBMS_OUTPUT.PUT_LINE('Salary of employee with ID 2 is: ' || v_salary);

    -- Check if value found (implicit cursor automatically does this)
    IF SQL%FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Record found.');
    ELSE
        DBMS_OUTPUT.PUT_LINE('No record found.');
    END IF;
END;
/
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> DECLARE
         v_salary employee.salary%TYPE;
     BEGIN
         -- Implicit cursor: single row query...
Show more...


Salary of employee with ID 2 is: 32000
Record found.
```

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Step 5 – Process Records Using Cursor Attributes

```
--------------------------------------------------------
-- Step 4 - Explicit Cursor Example
DECLARE
    -- Declare explicit cursor
    CURSOR emp_cursor IS
        SELECT id, name, department, salary FROM employee;

    -- Variables to hold fetched data
    v_id employee.id%TYPE;
    v_name employee.name%TYPE;
    v_department employee.department%TYPE;
    v_salary employee.salary%TYPE;
BEGIN
    OPEN emp_cursor; -- Open cursor
    LOOP
        FETCH emp_cursor INTO v_id, v_name, v_department, v_salary;
        EXIT WHEN emp_cursor%NOTFOUND; -- Exit when no more rows
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
PL/SQL procedure successfully completed.

Elapsed: 00:00:00.079

SQL> DECLARE
        -- Declare explicit cursor
        CURSOR emp_cursor IS
            SELECT id, name, department, salary FROM employee;...
Show more...
```

Learning Outcome

After completing this experiment, the learner will be able to:

- Understand the role of cursors in PL/SQL for handling multi-row query results.
- Differentiate between implicit cursors and explicit cursors.
- Use cursor attributes such as %FOUND, %NOTFOUND, %ROWCOUNT, and %ISOPEN.
- Develop PL/SQL programs that process database records row by row.
- Apply cursor-based logic in real-world business scenarios used in organizations like Oracle, Accenture, Scaler, and D. E. Shaw & Co.